

DATABASE SYSTEMS

Chap 3~ dbms_manipulation: RELATION ALGEBRA

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4.1.RELATIONAL ALGEBRA

- Is a **theoretical language** with operations that work on one or more relations to define another relation without changing the original relation
- Knowledge about relational algebra allows us to understand query execution and optimization in DBMs
- Relational database systems are expected to be equipped with a query language that can assist its users to query the database instances.

4.1.RELATIONAL ALGEBRA

- Relational algebra is a procedural query language, which takes instances of relations as input and yields instances of relations as output.
- It uses operators to perform queries.
- An operator can be either **unary** or **binary**. They accept relations as their input and yield relations as their output.

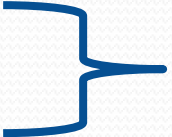
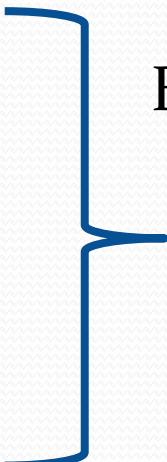
➤ Set operations

Unary and Binary operations

- Unary involves one operand (**relation**), whereas binary operation involves two operands (**relation**).
- Selection and projection are unary operations
- Union, difference, Cartesian product, and Join operations are binary operations

4.2 Relational Algebra Operations

- Relational algebra is performed recursively on a relation and intermediate results are also considered relations. The fundamental operations of relational algebra are as follows:

- Select
 - Project
- 
- Unary operation
- Union
 - Set different
 - Cartesian product
 - Rename
 - Intersection
- 
- Binary operation

SELECT OPERATION (σ)

- It selects tuples that satisfy the given predicate from a relation.
- Can be considered as *row wise filtering*
- **Notation** – $\sigma_p(r)$.
- Where σ stands for selection predicate and r stands for relation. p is propositional logic formula which may use connectors like **and**, **or**, and **not**. These terms may use relational operators like – =, $\neq$, $\geq$, $<$, $>$, $\leq$.



Examples1

1. $\sigma_{subject = "database"}(\text{Books})$

Output – Selects tuples from books where subject is 'database'.

2. $\sigma_{subject = "database" \text{ and } price = "450"}(\text{Books})$

Output – Selects tuples from books where subject is 'database' and 'price' is 450.

3. $\sigma_{subject = "database" \text{ and } price = "450" \text{ or } year > "2010"}(\text{Books})$

Output – Selects tuples from books where subject is 'database' and 'price' is 450 or those books published after 2010.

Example2

- Example , consider the STUDENT shown below

STUDENT		
Student Roll. No	Name	GPA
001	Aravind	7.2
002	Anand	7.5
003	Balu	8.2
004	Chitra	8.0
005	Deepa	8.5
006	Govind	7.2
007	Hari	6.5

- List the Roll.no, Name, and GPA of those students who are having GPA of above 8.0
- Query expressed in relational algebra as $\sigma_{GPA \succ 8} (Student)$

Student Roll. No	Name	GPA
003	Balu	8.2
005	Deepa	8.5

PROJECT OPERATION (Π)

It used to extract column(s) from a relation.

- **Notation** – $\Pi_{A_1, A_2, A_n} (r)$

Where A_1, A_2, A_n are attribute names of relation r .

Duplicate rows are automatically eliminated, as relation is a set.

- It's considered as a **column wise filtering**



Example

- Produce the list of salaries for all showing only the name and salary detail

$\Pi_{Name, salary}(Staff)$

STAFF				
Staff No	Name	Gender	Date of birth	Salary
SL21	Raghavan	M	1-5-76	15,000
SL22	Raghu	M	1-5-77	12,000
SL55	Babu	M	1-6-76	12,500
SL66	Kingsly	M	1-8-78	10,000

Output for the query

Name	Salary
Raghavan	15,000
Raghu	12,000
Babu	12,500
Kingsly	10,000

UNION OPERATION (U)

It performs binary union between two given relations and is defined as: $r \cup s = \{ t \mid t \in r \text{ or } t \in s \}$

- **Notion** – $r \cup s$

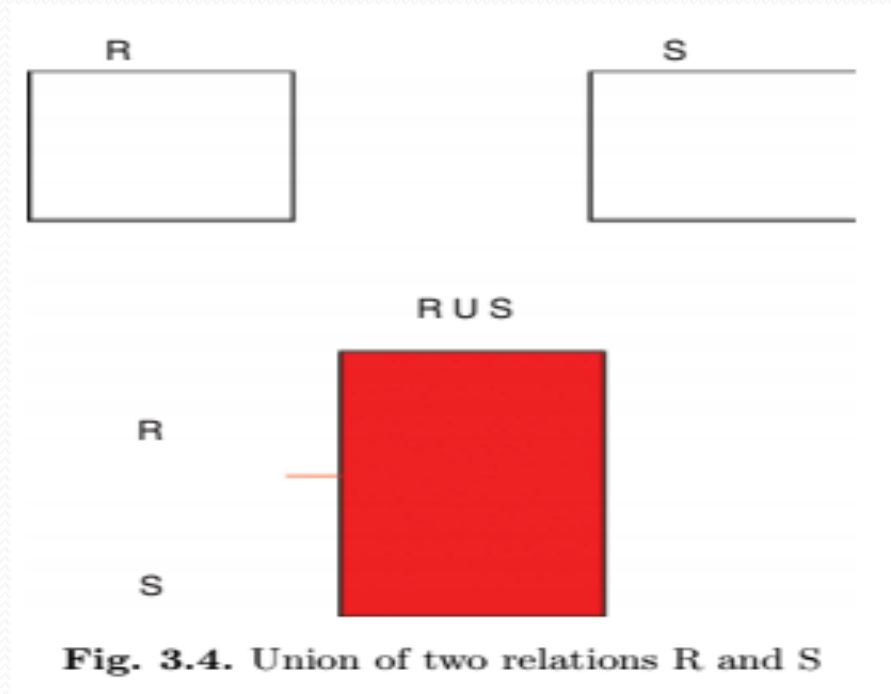
Where r and s are either database relations or relation result set (temporary relation).

- For a union operation to be valid, the following conditions must hold: r , and s must have the same number of attributes.
- Attribute domains must be compatible. Duplicate tuples are automatically eliminated.

EXAMPLE

$\Pi_{\text{author}} (\text{Books}) \cup \Pi_{\text{author}} (\text{Articles})$

Output – Projects the names of the authors who have either written a book or an article or both.



Example(BooksUArticles)

Books

Author_ID	Name
201	J.K. Rowling
202	George Orwell
203	Jane Austen

Articles

Author_ID	Name
203	Jane Austen
204	Malcolm Gladwell
205	Yuval Noah Harari

Result BooksUArticles

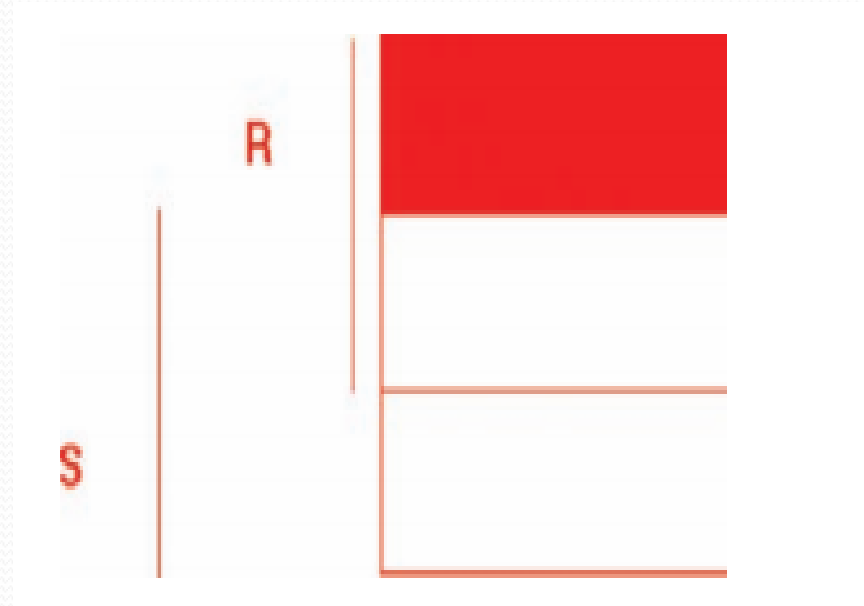
Author_ID	Name
201	J.K. Rowling
202	George Orwell
203	Jane Austen
204	Malcolm Gladwell
205	Yuval Noah Harari

SET DIFFERENCE (-)

The result of set difference operation is tuples, which are present in one relation but are not in the second relation.

- **Notation – $r - s$**

Finds all the tuples that are present in r but not in s .



EXAMPLE

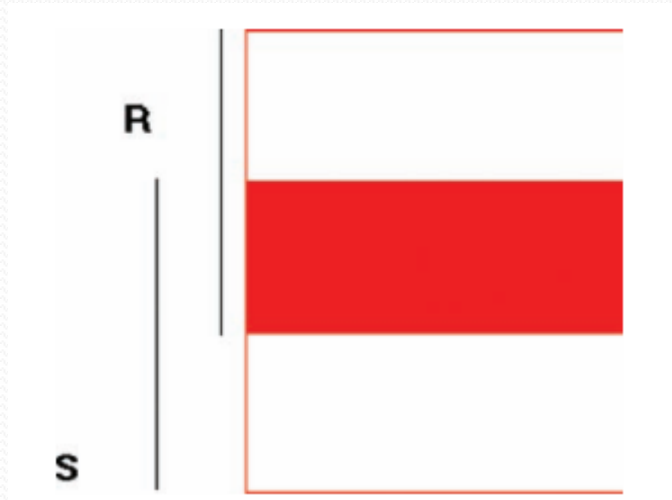
$\Pi_{\text{author}}(\text{Books}) - \Pi_{\text{author}}(\text{Articles})$

Output – Provides the name of authors who have written books but not articles.

4.2 Relational Algebra Operations

- **Intersection Operation**

- Defines a relation consisting of the set of all tuples that are in both R and S.
- The intersection of two relations R and S is denoted by $R \cap S$
- Its illustration is shown below



- From the shown relations

Example

$\Pi_{\text{author}}(\text{Books}) \cap \Pi_{\text{author}}(\text{Articles})$

Output: it will return the same authors who are in Books and who are in Articles.

- **Cartesian Product Operation**

- It defines a relation that is the concatenation of every tuples of relation R with every tuples of relation S. The result of Cartesian product contains all attributes from both relations R and S.
- Cartesian product between the two relations R and S is denoted by $R \times S$
- Example for illustration is shown below

R	S
a	1
b	2
	3

- Determine $R \times S$

R	S
a	1
a	2
a	3
b	1
b	2
b	3

Note:

No. of tuples in $R \times S = 2 * 3 = 6$

No. of attributes in $R \times S = 2$

EXAMPLE

$\Pi_{\text{author} = \text{'Gatera'}}(\text{Books X Articles})$

Output – Yields a relation, which shows all the books and articles written by **Gatera**.

RENAME OPERATION (ρ)

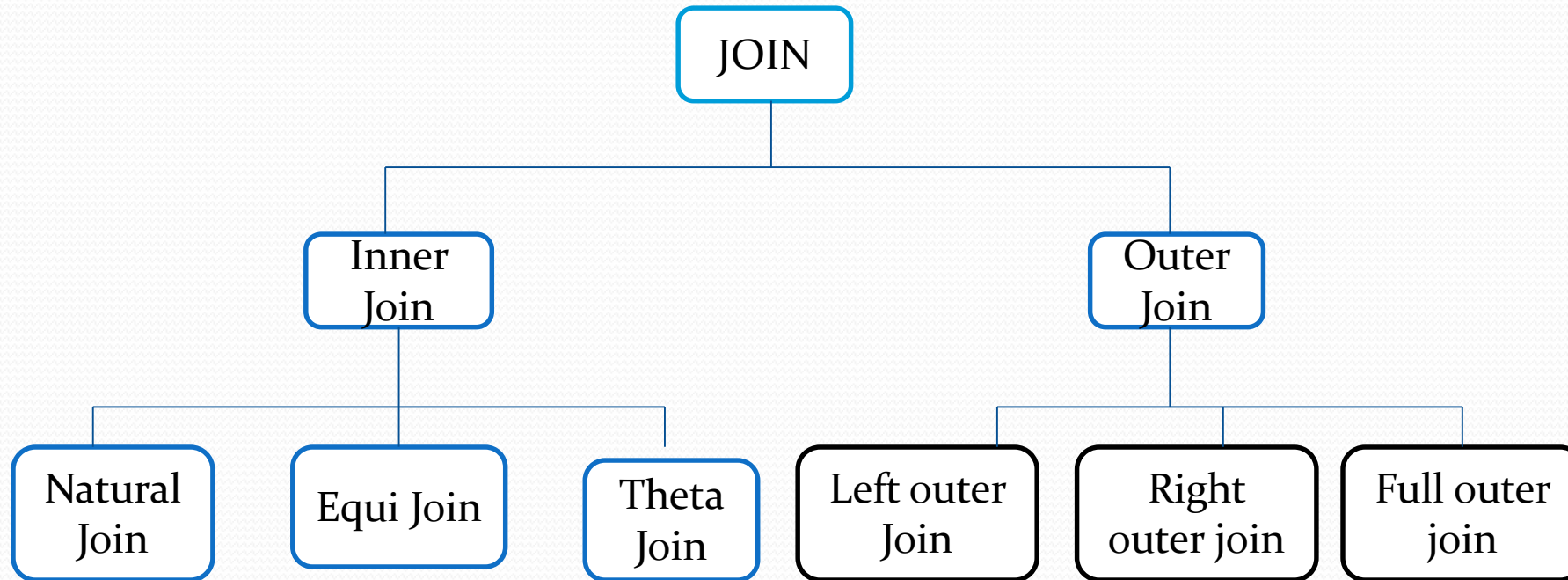
- The results of relational algebra are also relations but without any name. The rename operation allows us to rename the output relation.
- 'rename' operation is denoted with small Greek letter **rho** ρ .
- **Notation** – $\rho_x(E)$

Where the result of expression E is saved with name of x.

Relational Algebra Operations

Join Operations

- The most used operator in the relational algebra.
- Allows us to establish connections among data in different relations, taking advantage of the "valuebased" nature of the relational model



Join Operations

- Join operation combines two relations to form a new relation.
- The tables should be joined based on a common column and condition.
- The common column should be compatible in terms of domain.

Relational Algebra Operations

a. Natural Join

- ✓ The standard definition of natural join requires that the two join attributes, or each pair of corresponding join attributes, have the **same name** and **domain** in both relations.
- ✓ If this is not the case, a renaming operation is applied first.
- ✓ Acts on those matching attributes where the values of attributes in both relation is same.

r_1

Employee	Department
Smith	sales
Black	production
White	production

r_2

Department	Head
production	Mori
sales	Brown

$r_1 \bowtie r_2$

Employee	Department	Head
Smith	sales	Brown
Black	production	Mori
White	production	Mori

Joins can be Empty

r_1

Employee	Department
Smith	sales
Black	production
White	production

r_2

Department	Head
marketing	Mori
purchasing	Brown

$r_1 \bowtie r_2$

Employee	Department	Head

4.2. Relational Algebra Operations

b. Equi Join

- ✓ The most common use of join involves join conditions with equality comparisons only.
- ✓ Such a join, where the only comparison operator used is =, is called an EQUIJOIN.
- ✓ In the result of an EQUIJOIN we always have one or more pairs of attributes (whose names need not be identical) that have *identical values* in every tuple.
- ✓ The JOIN seen in the previous example was EQUIJOIN.

- ✓ Example

STAFF			DEPT	
Staff No	Job	Dept	Dept	Name
1	salesman	100	100	marketing
2	draftsman	101	101	civil

- ✓ Answer for the earlier query is equi-join of STAFF and DEPT(

- $$\text{EquiJoin } A \bowtie_{A.ID=B.ID} B$$

STAFF EQUI JOIN DEPARTMENT				
Staff No	Job	dept	dept	Name
1	salesman	100	100	marketing
2	draftsman	101	101	civil

4.2. Relational Algebra Operations

c. Theta Join

- ✓ A conditional join in which we impose condition other than equality condition
- ✓ If equality condition is imposed, then theta join become equi join.
- ✓ The symbol stands for the comparison operators: <, >, >=, <=.

$$\sigma_{\theta}(R \times S) \quad \text{or} \quad R1 \bowtie_{\theta} R2$$

Employees

Employee	Project
Smith	A
Black	A
Black	B

Projects

Code	Name
A	Venus
B	Mars

Employees $\bowtie_{\text{Project=Code}}$ Projects

Employee	Project	Code	Name
Smith	A	A	Venus
Black	A	A	Venus
Black	B	B	Mars

4.2. Relational Algebra Operations

d. Outer Join

- ✓ Matched pairs are retained unmatched values in other tables are left null

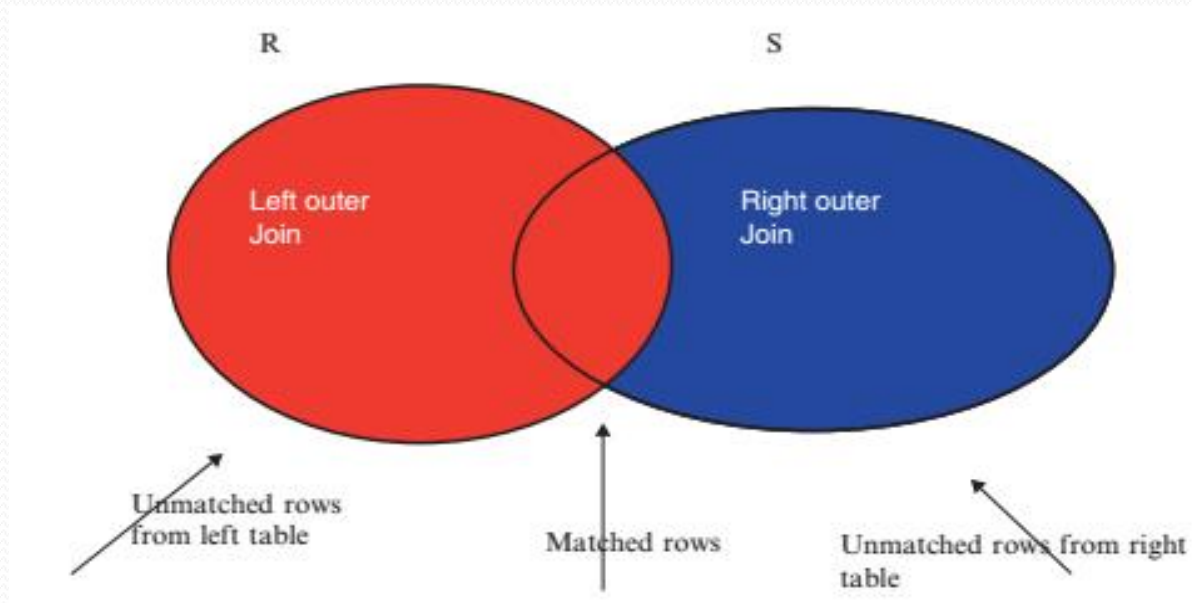
1. Full outer Join



2. Left outer Join



3. Right outer Join



4.2. Relational Algebra Operations

Outer Join

PEOPLE ⋈ **PEOPLE.Food = MENU.Food** **MENU**

Name	Age	People.Food	Menu.Food	Day
Raja	21	Idly	Idly	Tuesday
Ravi	22	Dosa	Dosa	Wednesday
Rani	20	Pizza	NULL	NULL
Devi	21	Pongal	Pongal	Monday

Table 3.2. Right outer join of PEOPLE and MENU relation

PEOPLE ⋈_r **PEOPLE.Food = Menu.Food** **MENU**

Name	Age	People.Food	Menu.Food	Day
Devi	21	Pongal	Pongal	Monday
Raja	21	Idly	Idly	Tuesday
Ravi	22	Dosa	Dosa	Wednesday
NULL	NULL	NULL	Fried rice	Thursday
NULL	NULL	NULL	Parotta	Friday

PEOPLE		
Name	Age	Food
Raja	21	Idly
Ravi	22	Dosa
Rani	20	Pizza
Devi	21	Pongal

MENU	
Food	Day
Pongal	Monday
Idly	Tuesday
Dosa	Wednesday
Fried rice	Thursday
Parotta	Friday

d. Left outer Join

All tuples of left relation R, are included in the resulting relation and if there exists tuples in R without any matching tuple in S then put NULL in S's attributes

e. Right outer Join

All tuples of the Right relation S, are included in the resulting relation and if there exists tuples in S without any matching in R, then put NULL in R's attributes

4.2. Relational Algebra Operations

f. Full outer Join

- ✓ All tuples of both participating relations are included in the resulting relation and if there is no matching tuples for both relations, their respective unmatched attributes are made NULL

Left	
A	B
100	Database
101	Mechanics
102	Electronics

[Table: Left Relation]

Right	
A	B
100	Alex
102	Maya
104	Mira

[Table: Right Relation]

Courses		HoD	
A	B	C	D
100	Database	100	Alex
101	Mechanics	---	---
102	Electronics	102	Maya
---	---	104	Mira

[Table: Full outer join output]

Limitations of Relational Algebra

- ✓ It can't do arithmetic
- ✓ It can not sort or print results in various formats
- ✓ It can not perform aggregates
- ✓ It can not modify the database
- ✓ It can not compute transitive closure



- END